

UNIT-1 Questions

1 Mark

Level 1

1. What is software testing?
2. Define error in software testing.
3. What is a fault?
4. What is a defect?
5. What is a failure in software?
6. What is the main objective of testing?
7. What does the V-Model represent?
8. What is test execution?
9. What is test automation?
10. What are test tools?

Level 2

1. Why is software testing needed?
2. What is the difference between an error and a defect?
3. How is a fault different from a failure?
4. Why can testing not prove software is completely bug-free?
5. Why is monitoring test execution important?
6. How does automation help in testing?
7. Why is the V-Model called a verification and validation model?
8. How do test tools improve testing efficiency?
9. What is meant by measuring test execution?
10. Why are testing activities performed in stages?

2 Mark

Level 2

1. Explain the need for software testing.
2. Describe the relationship between error, fault, defect, and failure.
3. Explain the objectives of software testing.
4. Describe the central issue in testing.
5. Explain why exhaustive testing is impossible.
6. Describe the role of monitoring during test execution.

Level 3

1. Classify the following as error, fault, defect, or failure:
A programmer writes a wrong formula in code.
2. A login page crashes when the wrong password is entered. Identify the failure and possible fault.
3. Suggest a situation where test automation would be more useful than manual testing.
4. Apply the V-Model to show which testing phase corresponds to system design.
5. Given a project with frequent builds, which testing approach (manual/automated) would you apply and why?

4 Marks

Level 3

1. Apply the V-Model to explain how development phases are linked with testing phases.
2. Illustrate with an example how an error leads to a failure in software.
3. Show how monitoring and measurement can improve test execution quality.
4. Demonstrate how test automation can be integrated into a testing lifecycle.

Level 4

1. Differentiate between error, fault, defect, and failure with suitable examples.
2. Analyze the advantages and limitations of test automation.
3. Compare manual testing and automated testing in terms of cost, effort, and reliability.
4. Analyze the limitations of software testing and explain why testing alone cannot ensure quality.
5. Examine the V-Model and analyze its strengths and weaknesses in real-world projects.

UNIT-2 Questions

1 Mark

Level 1

1. What is unit testing?
2. Define static unit testing.
3. What is dynamic unit testing?
4. What is mutation testing?
5. What is debugging?
6. What is a test case in unit testing?
7. What is a test driver?
8. What is a test stub?
9. Name any one unit testing tool.

Level 2

1. Why is unit testing important in software development?
2. How is static unit testing different from dynamic unit testing?
3. Why is mutation testing used?
4. How does unit testing help in defect prevention?
5. Why is debugging needed after testing?
6. How do stubs help in unit testing?
7. Why are drivers required in unit testing?
8. How do unit testing tools improve productivity?
9. Why is early defect detection important?
10. How does unit testing improve software quality?

2 Mark

Level 2

1. Explain the concept of unit testing.
2. Describe static unit testing with examples.
3. Explain dynamic unit testing.
4. Describe the role of defect prevention in unit testing.
5. Explain the need for debugging in software development.
6. Describe mutation testing in brief.

Level 3

1. Identify whether the following is static or dynamic unit testing:
Reviewing source code for logical errors.
2. Suggest how stubs would be used when testing a function that depends on a database.
3. Apply mutation testing to check the correctness of a conditional statement.
4. Given a module with multiple function calls, show how drivers can help in testing.
5. Demonstrate how unit testing can prevent defects from reaching later stages.
6. Apply debugging steps to fix a function that produces incorrect output.

4 Marks

Level 3

1. Illustrate the process of dynamic unit testing with a suitable example.
2. Show how stubs and drivers are used together during unit testing.
3. Demonstrate the steps involved in mutation testing with an example.
4. Apply defect prevention techniques during the unit testing phase.

Level 4

1. Differentiate between static and dynamic unit testing with advantages and limitations.
2. Analyze how mutation testing improves the effectiveness of test cases.
3. Compare unit testing and debugging in terms of purpose and process.
4. Analyze the role of unit testing tools in improving test coverage and efficiency.
5. Examine the impact of effective unit testing on overall software quality.